

Gyeongbin Ryoo

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EDUCATION

Dartmouth College Master of Science in Computer Science	Expected Graduation: 2027.06
Sungshin Women’s University Bachelor of Engineering in School of AI Convergence	Graduation: 2025.02 GPA: 4.07/4.5

PROJECTS

Hyundai NGV Undergraduate Researcher (Paid Position, Affiliated with Hyundai Motors)	2024.06.05 – 2025.01.31
- Contributed on the Industry-academia Collaboration Project focused on enhancing business models and integration scenarios for robot-friendly buildings - Developed technical standards, assisted in patent applications, and simulated optimal operational scenarios for robot delivery and parking services - Technical Stack: Utilized proprietary technologies (Confidential as per enterprise policy)	
Ministry of Science and ICT Undergraduate Researcher (Paid Position)	2023.03.01 – 2024.02.29
- Contributed on the Industry academia Collaboration Project focused on developing small-scale marine mobility solutions for collecting and recycling coastal floating waste - Analyzed ship movements as time series data and assisted in developing AI algorithms based on the analysis - Technical Stack: Python, Pandas, NumPy, PyTorch	
Hyundai Heavy Machinery Undergraduate Researcher (Paid Position)	2023.03.01 – 2023.11.30
- Contributed on the Industry academia Collaboration Project focused on ‘Development of Location-based Sloshing Load Prediction Program using Deep Learning’ - Responsible for building a login feature using AWS implementing functionality for administrators to upload AI models, and visualizing real-time AI model prediction results. - Technical Stack: C#, Python, PyTorch, AWS	

RESEARCH EXPERIENCE

Mobility Service Research Lab

Dec 2023 – Jan 2025

Undergraduate Research Assistant

- Researched real-time 3D scene reconstruction and freespace segmentation using NVIDIA's Nvblox and Isaac ROS libraries, focusing on voxel-based mapping, TSDF/ESDF distance fields for path planning, and local occupancy grid generation to support autonomous robotic navigation.
- Leveraged data from the NVIDIA Omniverse Isaac Sim metaverse application to enhance AI model performance through fine-tuning and validated the effectiveness of metaverse-generated data.

Marine Hydrodynamics AI Research Lab

Mar 2023 – Dec 2023

Undergraduate Research Assistant

- Conducted data preprocessing, including scaling and augmenting numerical datasets, to analyze correlations between a ship's 6-DoF motions and sloshing impact loads using experimental data from SNU Lab.
- Applied machine learning techniques, such as LightGBM and GAMs, to identify key motion parameters affecting sloshing loads and enhance predictive accuracy.
 - Contributed to a UROP project by developing and implementing deep learning software

HONORS & AWARDS

Top Academic Achievement Scholarship (90% of tuition)	2024.08.21
UROP Scholarship	2023.09.11
Sungshin Scholarship (Academic Excellence)	2023.08.22
Academic Excellence1 Scholarship (80% of tuition)	2023.02.20
Broadcast Journalism Scholarship A/B/C	2022.12 ~ 2024.07
Improvement of Language Skills Scholarship A / B (English)	2022.02.03
KICS Summer Conference Undergraduate Excellent Paper Award (Encouragement)	2024.06.19
AI Convergence Department Software Development Competition (Encouragement)	2023.09.04

PAPERS

Conference

1. "A Fine-Tuning Approach for Object Detection Models Using Virtually Generated Data," presented at ICTC 2024 (Oct 16-18, 2024). (Co-first author)
2. "Simulation-Based Service Scenarios Analysis for Commercial Deployment: Robot-Friendly Building Case," accepted at KOSIME Domestic Conference, July 2024. (Co-author)
3. "Performance Analysis of Semantic Segmentation in Fine-Tuned Models Based on Virtual Environment," presented at KICS Symposium, June 2024. (First author)
4. "Analyzing Experimental Data of 6-DOF Motions and Sloshing Impacts in LNG Carriers," published in ASME OMAE 2024 (June 2024). (Co-author)

EXTRACURRICULAR ACTIVITIES

Club Member

March 2024 – August 2024

ELECourse, Electronics Club

- Designed and optimized voice signal analysis hardware using LM386 circuits and Pspice simulations, incorporating noise reduction techniques and creating a user-friendly 3D-printed device for real-time voice processing.,
- Developed and fine-tuned a Wav2Vec2 AI model for voice emotion classification, achieving 82% accuracy through data preprocessing, augmentation, and FIR filter-based noise removal using Matlab and Python.
- Collaborated on a team project to create a depression diagnosis model, enhancing the performance of Pure CNN, Pure LSTM, and CNN-LSTM hybrid models, and achieving 90.4% accuracy by combining spatial and temporal feature extraction.

Editor-in-Chief

December 2022 – February 2025

Mirror, Sungshin Women's University English Newspaper Media

- Oversaw the university press operations, including magazine publication, inter-department communication, and budget management
- Wrote as the dedicated journalist for the Science and Technology corner, focusing on AI and Robotics.
- Led the transition from print to digital, preventing shutdown by addressing declining readership, and managed the digital newspaper website.

SKILLS

- English (Fluent - TOEFL 107)
- Programming: Python, C++, PyTorch, Node.js, AWS